

Computer Networks I

Multiplexing

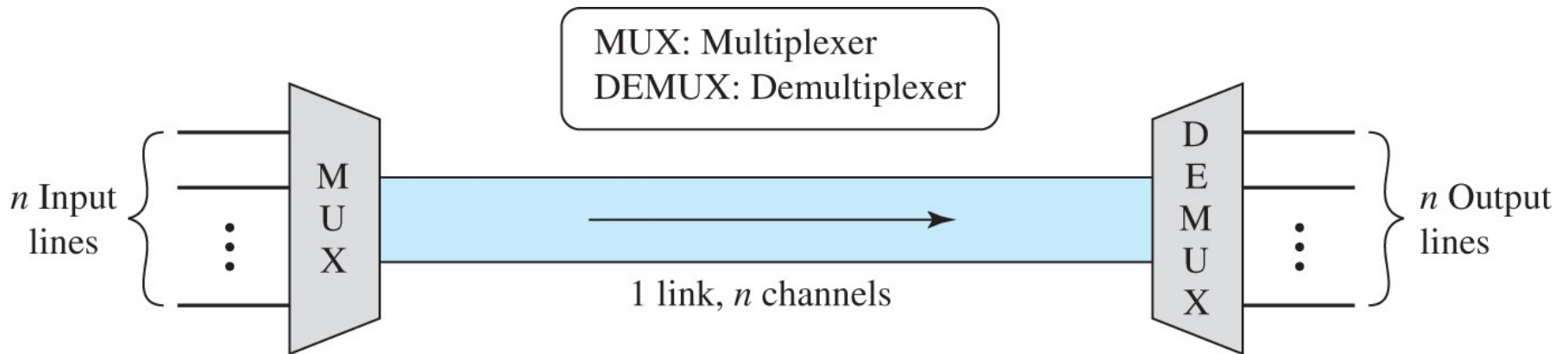
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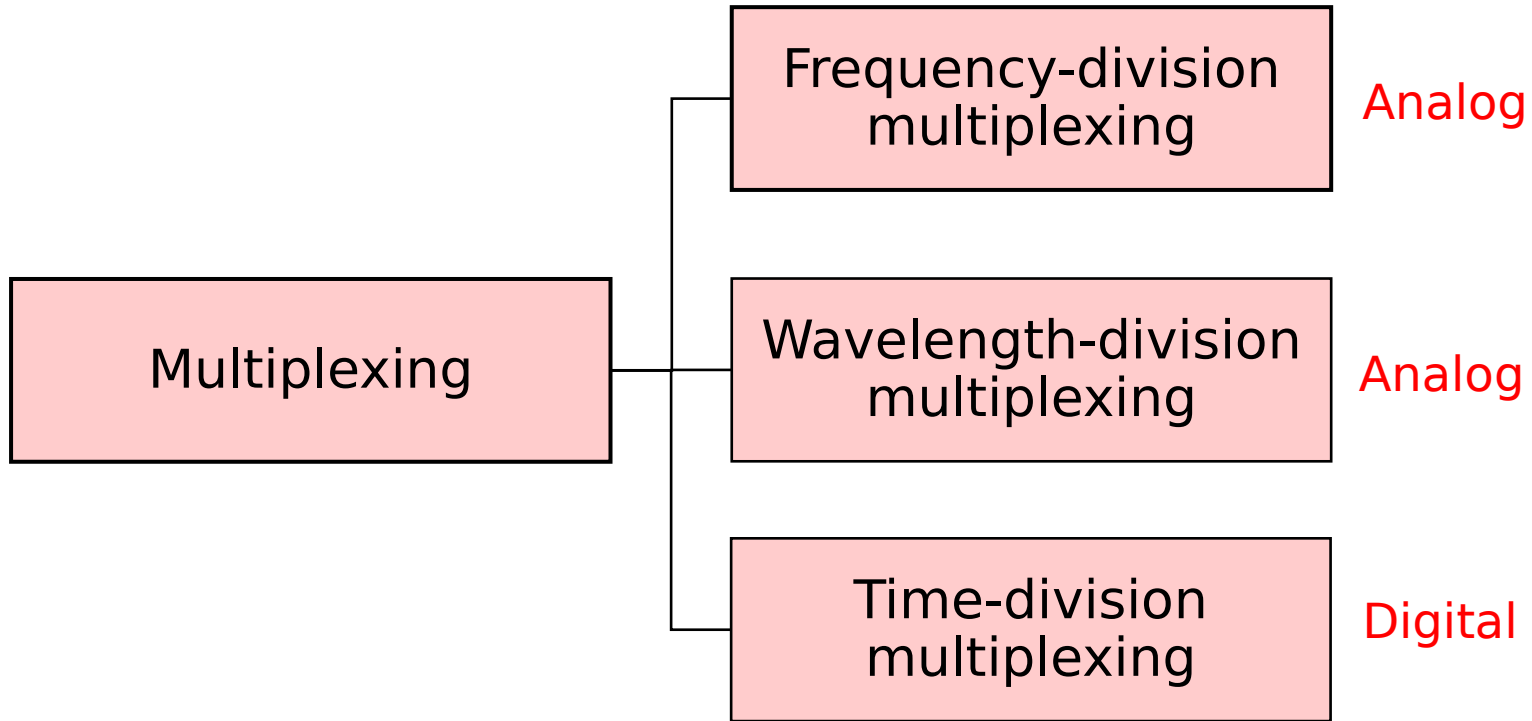
IIT Kanpur

Multiplexing

- ❑ Multiplexing is the set of techniques that allows the simultaneous transmission of multiple signals across a single data link
- ❑ **Multiplexer (MUX):** A device that combines several signals into a single signal
- ❑ **Demultiplexer (DEMUX):** A device that performs the inverse operation



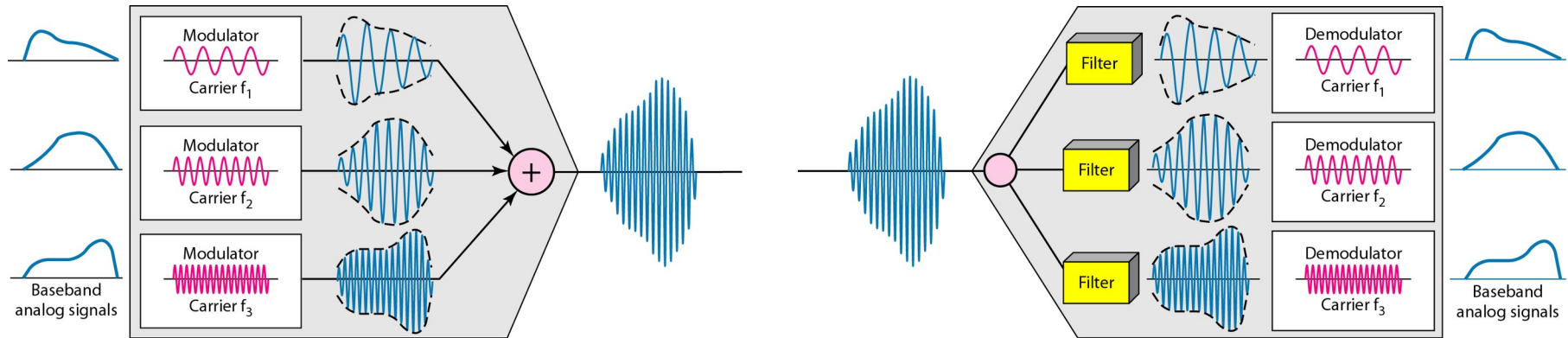
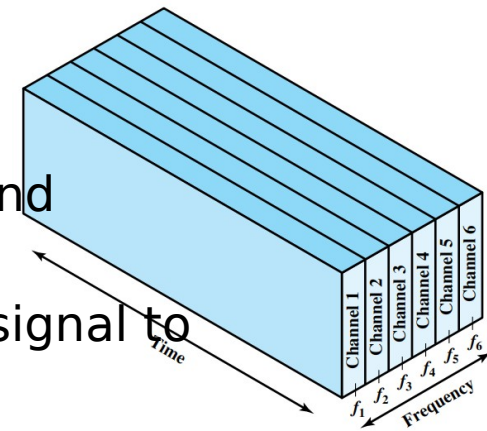
Multiplexing



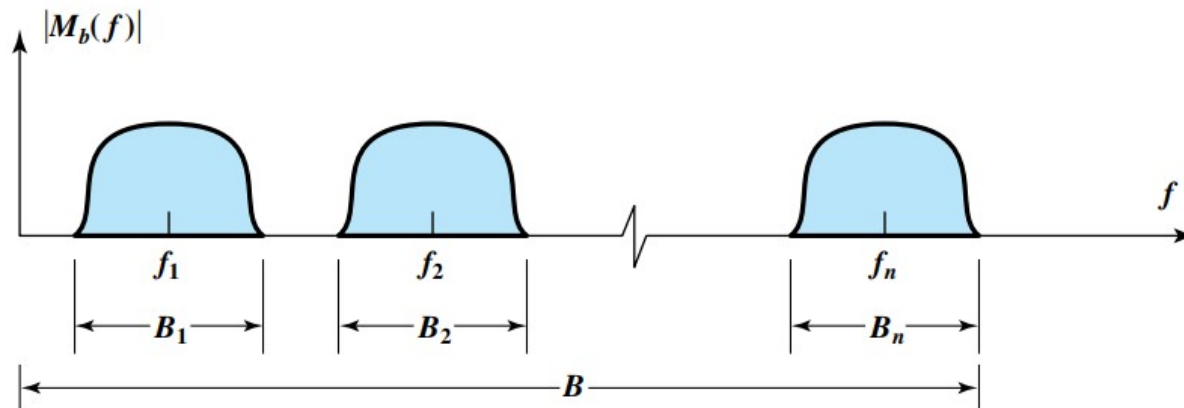
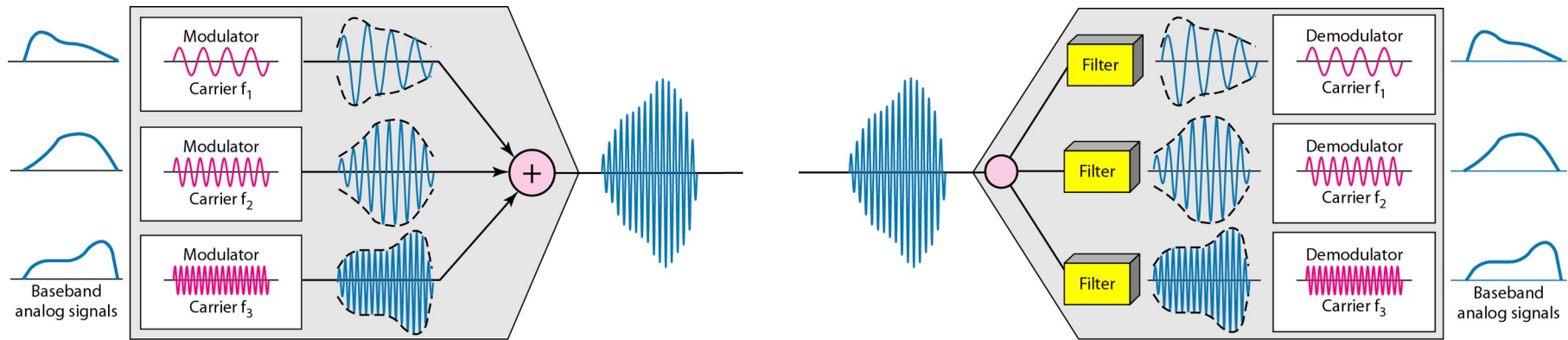
Frequency Division Multiplexing

FDM

- Each signal is allocated a different frequency band
- Usually used with analog signals
- Modulation equipment is needed to move each signal to the required frequency band (channel)
- Multiple carriers are used, each is called sub-carrier



FDM



FDM

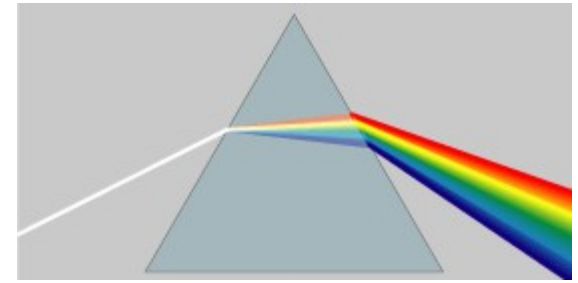
- ❑ Five channels, each with a 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between the channels to prevent interference?



Wavelength Division Multiplexing

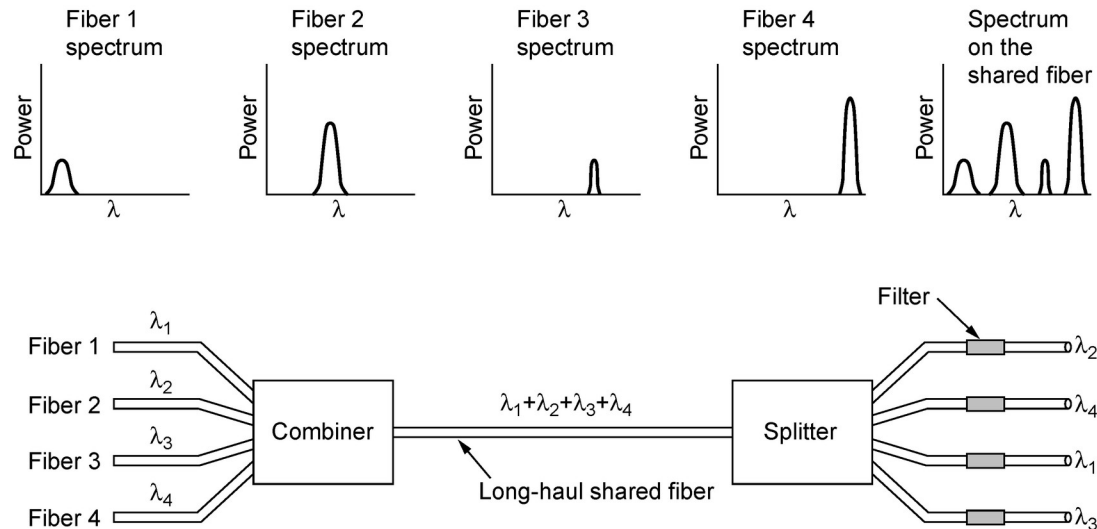
WDM

- ❑ WDM is conceptually the same as FDM
 - ❑ Multiplexing and demultiplexing involve light signals transmitted through fiber-optic channels



Src: https://commons.wikimedia.org/wiki/File:Prism_rainbow_schema.png

- ❑ Combining and splitting of light sources can be handled by a prism



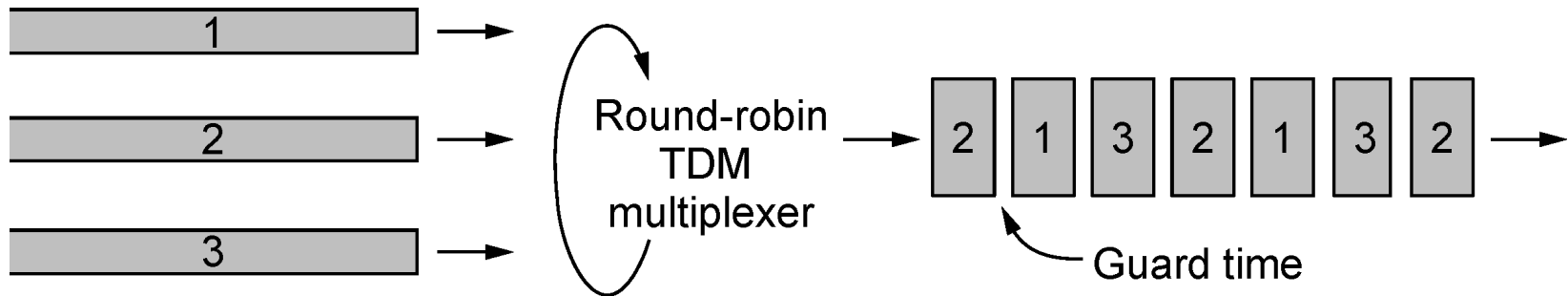
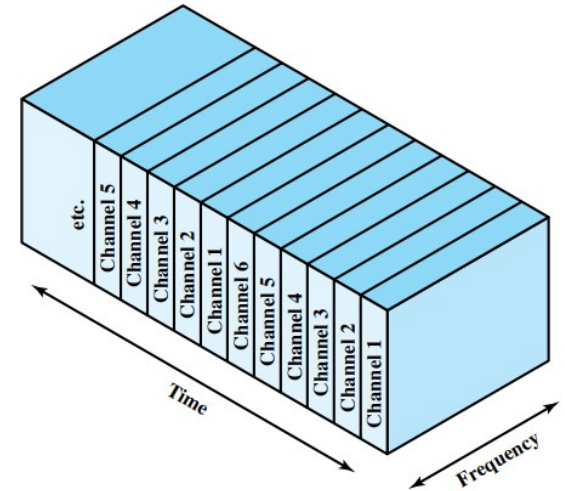
Time Division Multiplexing

TDM

- User takes turns
 - Each one periodically get the entire bandwidth

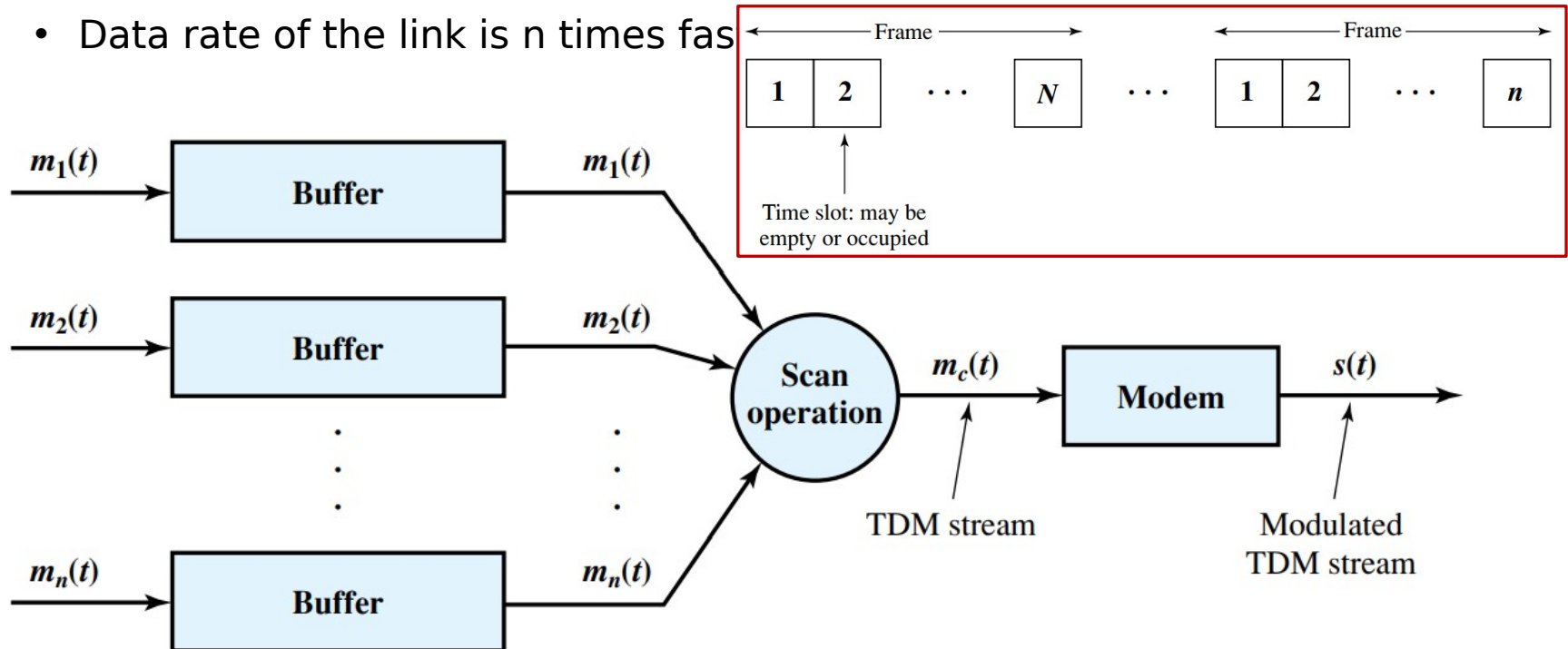
□ TDM:

- Synchronous TDM
- Asynchronous TDM



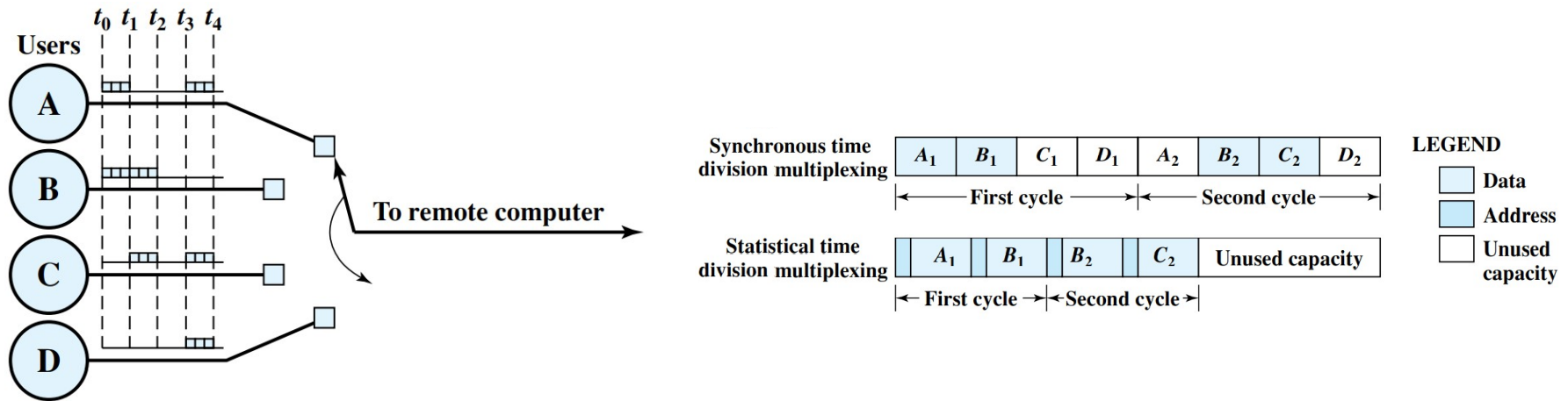
Synchronous TDM

- Multiplexer allocates exactly the same time slot to each device at all times, whether or not a device has anything to transmit
 - Data rate of the link is n times faster



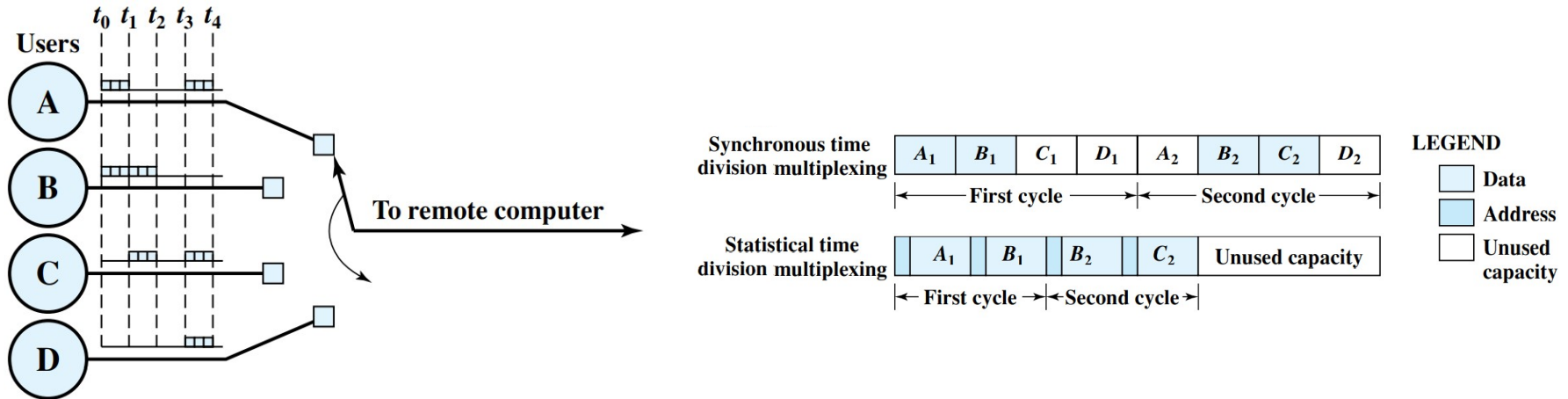
Statistical TDM

- In synchronous TDM it is often the case that many of the time slots in a frame are **wasted**
- Asynchronous time-division multiplexing, or statistical time-division multiplexing, is designed to avoid this type of waste



Statistical TDM

- Line data rate can be lower than input lines rates
- Overhead per slot for statistical TDM because each slot carries an address as well as data
- May have problems during **peak periods**
 - Must buffer inputs



Code Division Multiple Access

CDMA

$$A = (-1 \ -1 \ -1 \ +1 \ +1 \ -1 \ +1 \ +1)$$

$$B = (-1 \ -1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1)$$

$$C = (-1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1 \ -1)$$

$$D = (-1 \ +1 \ -1 \ -1 \ -1 \ -1 \ +1 \ -1)$$

All chip sequences are orthogonal

$$S \bullet T = \frac{1}{m} \sum_{i=1}^m S_i T_i = 0 \qquad S \bullet \bar{T} = 0$$

$$S \bullet S = \frac{1}{m} \sum_{i=1}^m S_i S_i = \frac{1}{m} \sum_{i=1}^m S_i^2 = \frac{1}{m} \sum_{i=1}^m (\pm 1)^2 = 1$$

CDMA

$$A = (-1 \ -1 \ -1 \ +1 \ +1 \ -1 \ +1 \ +1)$$

$$B = (-1 \ -1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1)$$

$$C = (-1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1 \ -1)$$

$$D = (-1 \ +1 \ -1 \ -1 \ -1 \ -1 \ +1 \ -1)$$

$$X = A + \bar{B} + C = (-1 \ +1 \ -3 \ +3 \ +1 \ -1 \ -1 \ +1)$$

CDMA

$$A = (-1 \ -1 \ -1 \ +1 \ +1 \ -1 \ +1 \ +1)$$

$$B = (-1 \ -1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1)$$

$$C = (-1 \ +1 \ -1 \ +1 \ +1 \ +1 \ -1 \ -1)$$

$$D = (-1 \ +1 \ -1 \ -1 \ -1 \ -1 \ +1 \ -1)$$

$$X = A+B+\bar{C}+D = (-2 \ -2 \ 0 \ -2 \ 0 \ -2 \ +4 \ 0)$$

CDMA

□ Proof:

- Let's consider the case of $S = (A + \bar{B} + C)$

$$\begin{aligned} S \cdot C &= (A + \bar{B} + C) \cdot C = A \cdot C + \bar{B} \cdot C + C \cdot C \\ &= 0 + 0 + 1 = 1 \end{aligned}$$

- One significant assumption is that, the chips are synchronized in time at the receiver
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Summary

□ Different multiplexing techniques discussed:

- Frequency division multiplexing
 - Wavelength division multiplexing
 - Time division multiplexing
 - Synchronous TDM
 - Asynchronous TDM
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